

Plugging into the future : An exploration wise consumption patterns Dashboard

TEAM ID	SWTID-2026-7684
PROJECT TITLE	Plugging into the future: An exploration wise consumption pattern
MAX MARKS	5 MARKS

4.2 – Proposed Solution

1. Predictive Load Shifting Engine

- The Mechanism: An algorithm analyses dynamic utility pricing structures and weather forecasts hourly.
- The Action: It automatically schedules energy-intensive tasks (e.g., water heating, laundry, pool pumps) to run when electricity is cheapest and greenest.

2. Bidirectional EV & Battery Orchestration

- The Mechanism: Integrates Vehicle-to-Home (V2H) and localized battery storage systems.
- The Action: The system charges home and vehicle batteries during off-peak windows. It discharges that stored energy to power the home during high-cost peak grid periods, avoiding expensive rates entirely.

3. Automated "Phantom Load" Mitigation

- The Mechanism: Smart plugs and IoT relays monitor idle power consumption.
- The Action: Power is cut to standby devices (gaming consoles, desktop setups, media centres) during scheduled sleeping hours or when geofencing indicates the property is vacant.

4. Behavioural Gamification Engine

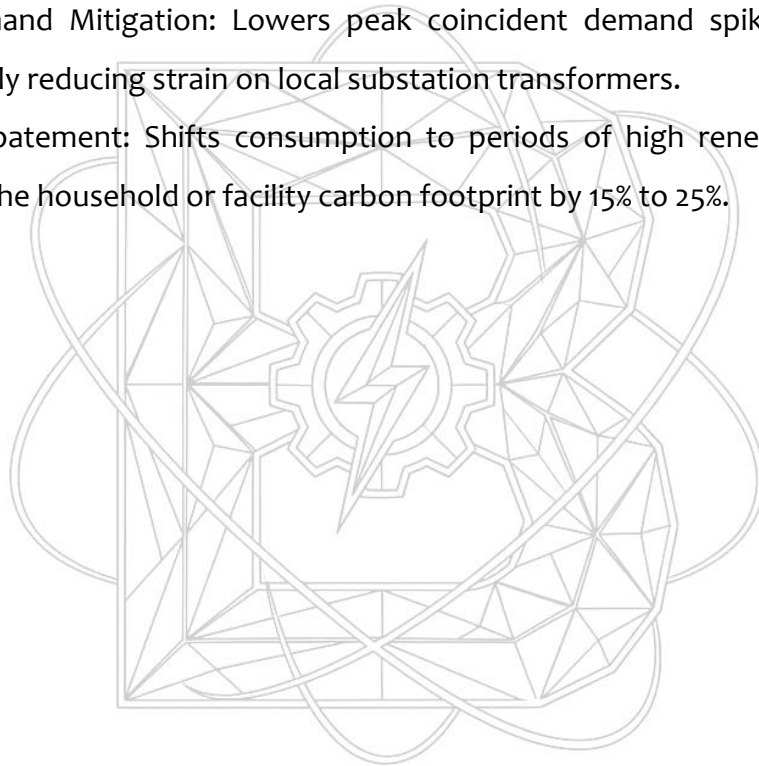
- The Mechanism: A mobile dashboard tracks real-time usage and pushes actionable alerts.

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- The Action: Users receive direct cash-back incentives, neighbourhood efficiency leaderboard rankings, and rewards for manually lowering consumption during emergency grid stress events.

Expected Strategic Outcomes

- Cost Reduction: Reduces monthly electricity bills by 20% to 35% via automated Time-of-Use optimization.
- Peak Demand Mitigation: Lowers peak coincident demand spikes by up to 40%, significantly reducing strain on local substation transformers.
- Carbon Abatement: Shifts consumption to periods of high renewable generation, lowering the household or facility carbon footprint by 15% to 25%.



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